

Solution of the System of Algebraic Equations by Matrix Method

Representation of a linear system: The general system of m equations and n unknowns can be written as

$$\begin{array}{ccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n = b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n = b_2 \\ \vdots & & \vdots & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \cdots & + & a_{mn}x_n = b_m \end{array}$$

Using matrix notation the system can be written as

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$Ax = b$$

The augmented coefficient matrix of the system is

$$\hat{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & & a_{2n} & b_2 \\ & \vdots & \ddots & \vdots & \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

Solution of the System of Algebraic Equations by Matrix Method

Solutions to equations

- a. A system of linear equations with coefficient matrix A which is $m \times n$, a right hand side vector b which is $m \times 1$, and augmented matrix $\hat{A}$ has a solution if and only if

$$\text{rank}(A) = \text{rank}(\hat{A})$$

This means that we can write the b vector as a linear combination of the columns of A .

- b. A linear system of equations must have either no solution, one solution, or infinitely many solutions.
- c. If the rank of $A = r = n < m$, the linear system $Ax = b$ has no solution or exactly one solution. In other words, if the matrix A is of full column rank and has less columns than rows, the system will be inconsistent, or will have exactly one solution.
- d. If the rank of $A = r = m < n$, the linear system $Ax = b$ will always have a solution and there will be an infinite number of solutions. In other words, if the matrix A is of full row rank and has more columns than rows, the system will always have an infinite number of solutions.
- e. If the rank of $A = r$ where $r < m$ and $r < n$, the linear system $Ax = b$ will either have no solutions or there will be an infinite number of solutions. In other words, if the matrix A has neither full row nor column rank, the system will be inconsistent or will have an infinite number of solutions.
- f. If the rank of $A = r = m = n$, the linear system $Ax = b$ will exactly one solution. In other words, if the matrix A is square and has full rank, the system will have a unique solution.

A coefficient matrix is said to be **nonsingular**, that is, the corresponding linear system has one and only one solution for every choice of right hand side $b_1, b_2, \dots, b_m$, if and only if

$$\text{number of rows of } A = \text{number of columns of } A = \text{rank}(A)$$

Solution of the System of Algebraic Equations by Matrix Method

1.5.1. *Example of a system with one solution.* Consider the system of equations

$$\begin{aligned}x_1 - x_2 - x_3 &= 1 \\ -3x_1 + 4x_2 + 2x_3 &= -8 \\ 2x_1 + x_2 - 4x_3 &= -10\end{aligned}\tag{27}$$

written in matrix form as

$$Ax = b$$
$$\begin{pmatrix} 1 & -1 & -1 \\ -3 & 4 & 2 \\ 2 & 1 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ -8 \\ -10 \end{pmatrix}\tag{28}$$

The augmented matrix is of the form

$$\tilde{A} = \left(\begin{array}{ccc|c} 1 & -1 & -1 & 1 \\ -3 & 4 & 2 & -8 \\ 2 & 1 & -4 & -10 \end{array} \right)\tag{29}$$

After appropriate row and column operations we obtain

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{array} \right)\tag{30}$$

Solution of the System of Algebraic Equations by Matrix Method

1.5.5. *Example of a system with three equations and three unknowns and infinite solutions.* Consider the system of equations

$$\begin{aligned}x_1 - x_2 - x_3 &= 1 \\ -3x_1 + 4x_2 + 2x_3 &= -8 \\ -2x_1 + 3x_2 + x_3 &= -7\end{aligned}\tag{46}$$

written in matrix form as

$$Ax = b$$
$$\begin{pmatrix} 1 & -1 & -1 \\ -3 & 4 & 2 \\ -2 & 3 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ -8 \\ -7 \end{pmatrix}\tag{47}$$

The augmented matrix is of the form

$$\tilde{A} = \left(\begin{array}{ccc|c} 1 & -1 & -1 & 1 \\ -3 & 4 & 2 & -8 \\ 2 & 3 & 1 & -7 \end{array} \right)\tag{48}$$

After appropriate row and column operations we obtain

$$\left(\begin{array}{ccc|c} 1 & 0 & -2 & -4 \\ 0 & 1 & -1 & -5 \\ 0 & 0 & 0 & 0 \end{array} \right)\tag{49}$$

Solution of the System of Algebraic Equations by Matrix Method

1.5.6. *Example of a system with three equations and three unknowns and no solution.* Consider the system of equations

$$\begin{aligned}x_1 - x_2 + 2x_3 &= 5 \\2x_1 + 3x_2 - x_3 &= 5 \\3x_1 + 2x_2 + x_3 &= 9\end{aligned}\tag{50}$$

written in matrix form as

$$Ax = b$$
$$\begin{pmatrix} 1 & -1 & 2 \\ 2 & 3 & -1 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \\ 9 \end{pmatrix}\tag{51}$$

The augmented matrix is of the form

$$\tilde{A} = \left(\begin{array}{ccc|c} 1 & -1 & 2 & 5 \\ 2 & 3 & -1 & 5 \\ 3 & 2 & 1 & 9 \end{array} \right)\tag{52}$$

After appropriate row and column operations we obtain

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & -4 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & -1 \end{array} \right)\tag{53}$$

Solution of the System of Algebraic Equations by Matrix Method

Example 1 Solve the following system of equations.

$$x_1 - 2x_2 + 3x_3 = -2$$

$$-x_1 + x_2 - 2x_3 = 3$$

$$2x_1 - x_2 + 3x_3 = 1$$

The augmented matrix of the system.

$$\begin{pmatrix} 1 & -2 & 3 & -2 \\ -1 & 1 & -2 & 3 \\ 2 & -1 & 3 & 1 \end{pmatrix} \xrightarrow[H_{31}(-2)]{H_{21}(1)} \begin{pmatrix} 1 & -2 & 3 & -2 \\ 0 & -1 & 1 & 1 \\ 0 & 3 & -3 & 5 \end{pmatrix} \xrightarrow{H_{32}(3)} \left(\begin{array}{ccc|c} 1 & -2 & 3 & -2 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & 0 & 8 \end{array} \right)$$

Here, Rank of A and augmented matrix $\hat{A}$ are not same. So the system is inconsistent and the system has no solution.

Example 2 Solve the following system of equations.

$$x_1 - 2x_2 + 3x_3 = -2$$

$$-x_1 + x_2 - 2x_3 = 3$$

$$2x_1 - x_2 + 3x_3 = -7$$

The augmented matrix of the system.

$$\begin{pmatrix} 1 & -2 & 3 & -2 \\ -1 & 1 & -2 & 3 \\ 2 & -1 & 3 & -7 \end{pmatrix} \xrightarrow[H_{31}(-2)]{H_{21}(1)} \begin{pmatrix} 1 & -2 & 3 & -2 \\ 0 & -1 & 1 & 1 \\ 0 & 3 & -3 & -3 \end{pmatrix} \xrightarrow{H_{32}(3)} \left(\begin{array}{ccc|c} 1 & -2 & 3 & -2 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right) \xrightarrow{H_{12}(-2)} \left(\begin{array}{ccc|c} 1 & 0 & 1 & -4 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Here, rank of A and augmented matrix $\hat{A}$ are same, and less than number unknown. So the system has infinite number of solutions.

Solution of the System of Algebraic Equations by Matrix Method

If we now go back to the equation we get the following equation

$$\begin{array}{l} x_1 + x_3 = -4 \\ -x_2 + x_3 = 1 \end{array} \quad \Rightarrow \quad \begin{array}{l} x_1 = -x_3 - 4 \\ x_2 = x_3 - 1 \end{array}$$

The solution can be written as $x_1 = -t - 4, x_2 = t - 1, x_3 = t$ $t = \text{any real number}$

Example 3 Solve the following system of equations.

$$-2x_1 + x_2 - x_3 = 4$$

$$x_1 + 2x_2 + 3x_3 = 13$$

$$3x_1 + x_3 = -1$$

The augmented matrix of the system.

$$\begin{pmatrix} -2 & 1 & -1 & 4 \\ 1 & 2 & 3 & 13 \\ 3 & 0 & 1 & -1 \end{pmatrix} \xrightarrow{H_{12}} \begin{pmatrix} 1 & 2 & 3 & 13 \\ -2 & 1 & -1 & 4 \\ 3 & 0 & 1 & -1 \end{pmatrix} \xrightarrow{H_{21}(2), H_{31}(-3)} \begin{pmatrix} 1 & 2 & 3 & 13 \\ 0 & 5 & 5 & 30 \\ 0 & -6 & -8 & -40 \end{pmatrix} \xrightarrow{H_2(\frac{1}{5})} \begin{pmatrix} 1 & 2 & 3 & 13 \\ 0 & 1 & 1 & 6 \\ 0 & -6 & -8 & -40 \end{pmatrix} \xrightarrow{H_{32}(6)} \begin{pmatrix} 1 & 2 & 3 & 13 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & -2 & -4 \end{pmatrix}$$

If we now go back to the equation we get the following equation

$$x_1 + 2x_2 + 3x_3 = 13$$

$$x_2 + x_3 = 6$$

$$-2x_3 = -4$$

The solution of this system of equation is $x_1 = -1, x_2 = 4, x_3 = 2$

Solution of the System of Algebraic Equations by Matrix Method

$$\text{Solve } \begin{cases} x_1 + x_2 + x_3 + x_4 = 0 \\ x_1 + 3x_2 + 2x_3 + 4x_4 = 0 \\ 2x_1 + x_3 - x_4 = 0 \end{cases}.$$

Solution:

$$\begin{aligned} &= \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 3 & 2 & 4 & 0 \\ 2 & 0 & 1 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 2 & 1 & 3 & 0 \\ 0 & -2 & -1 & -3 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 2 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{1}{2} & \frac{3}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & \frac{1}{2} & \frac{3}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Find all non-trivial solutions of:

$$(a) \begin{cases} x_1 - 2x_2 + 3x_3 = 0 \\ 2x_1 + 5x_2 + 6x_3 = 0 \end{cases}$$

$$(c) \begin{cases} x_1 + 2x_2 + 3x_3 = 0 \\ 2x_1 + x_2 + 3x_3 = 0 \\ 3x_1 + 2x_2 + x_3 = 0 \end{cases}$$

$$(b) \begin{cases} 2x_1 - x_2 + 3x_3 = 0 \\ 3x_1 + 2x_2 + x_3 = 0 \\ x_1 - 4x_2 + 5x_3 = 0 \end{cases}$$

$$(d) \begin{cases} 4x_1 - x_2 + 3x_3 + x_4 = 0 \\ 2x_1 + 3x_2 - x_3 - 2x_4 = 0 \\ 7x_2 - 4x_3 - 5x_4 = 0 \\ 2x_1 - 11x_2 + 7x_3 + 8x_4 = 0 \end{cases}$$

$$\begin{cases} x_1 + x_2 + x_3 + x_4 = 0 \\ x_1 + 3x_2 + 2x_3 + 4x_4 = 0 \\ 2x_1 + x_3 - x_4 = 0 \end{cases}.$$

Solution of the System of Algebraic Equations by Matrix Method

. Find all solutions of:

$$(a) \quad x_1 - 2x_2 + x_3 - 3x_4 = 1$$

$$(b) \quad \begin{cases} x_1 + x_2 + x_3 = 4 \\ 2x_1 + 5x_2 - 2x_3 = 3 \end{cases}$$

$$(c) \quad \begin{cases} x_1 + x_2 + x_3 = 4 \\ 2x_1 + 5x_2 - 2x_3 = 3 \\ x_1 + 7x_2 - 7x_3 = 5 \end{cases}$$

$$(d) \quad \begin{cases} x_1 + x_2 + x_3 + x_4 = 0 \\ x_1 + x_2 + x_3 - x_4 = 4 \\ x_1 + x_2 - x_3 + x_4 = -4 \\ x_1 - x_2 + x_3 + x_4 = 2 \end{cases}$$

$$\begin{cases} 2x_1 + x_2 + 5x_3 + x_4 = 5 \\ x_1 + x_2 - 3x_3 - 4x_4 = -1 \\ 3x_1 + 6x_2 - 2x_3 + x_4 = 8 \\ 2x_1 + 2x_2 + 2x_3 - 3x_4 = 2 \end{cases}$$

Solution of the System of Algebraic Equations by Matrix Method

Investigate for what values of λ and μ the system of equations have

(i) no solution (ii) unique solution (iii) infinite number of solution

$$\begin{aligned}x + y + z &= 6 \\x + 2y + 3z &= 10 \\x + 2y + \lambda z &= \mu\end{aligned}$$

Solution: In matrix form the above equation can be written as
$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 6 \\ 10 \\ \mu \end{pmatrix}$$

Here

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{pmatrix} \quad B = \begin{pmatrix} 6 \\ 10 \\ \mu \end{pmatrix} \quad X = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \hat{A} = \begin{pmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{pmatrix}$$

Reduce augmented matrix to the echelon form, we get

$$\begin{pmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{pmatrix} \xrightarrow[H_{31}(-1)]{H_{21}(-1)} \begin{pmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda - 1 & \mu - 6 \end{pmatrix} \xrightarrow{H_{32}(-1)} \begin{pmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda - 3 & \mu - 10 \end{pmatrix}$$

(i) no solution

$$\lambda - 3 = 0 \quad \mu - 10 \neq 0$$

(ii) unique solution

$$\lambda - 3 \neq 0 \quad \mu - 10 \neq 0$$

(iii) infinite number of solution

$$\lambda - 3 = 0 \quad \mu - 10 = 0$$

Eigenvalues and Eigenvectors

So, let's start with the following. If we multiply an $n \times n$ matrix by an $n \times 1$ vector we will get a new $n \times 1$ vector back. In other words,

$$A\vec{\eta} = \vec{y}$$

What we want to know is if it is possible for the following to happen. Instead of just getting a brand new vector out of the multiplication is it possible instead to get the following,

$$A\vec{\eta} = \lambda\vec{\eta} \tag{1}$$

In other words, is it possible, at least for certain λ and $\vec{\eta}$, to have matrix multiplication be the same as just multiplying the vector by a constant? Of course, we probably wouldn't be talking about this if the answer was no. So, it is possible for this to happen, however, it won't happen for just any value of λ or $\vec{\eta}$. If we do happen to have a λ and $\vec{\eta}$ for which this works (and they will always come in pairs) then we call λ an **eigenvalue** of A and $\vec{\eta}$ an **eigenvector** of A .

So, how do we go about finding the eigenvalues and eigenvectors for a matrix? Well first notice that if $\vec{\eta} = \vec{0}$ then (1) is going to be true for any value of λ and so we are going to make the assumption that $\vec{\eta} \neq \vec{0}$. With that out of the way let's rewrite (1) a little.

$$A\vec{\eta} - \lambda\vec{\eta} = \vec{0}$$

$$A\vec{\eta} - \lambda I_n \vec{\eta} = \vec{0}$$

$$(A - \lambda I_n)\vec{\eta} = \vec{0}$$

Eigenvalues and Eigenvectors

Example1: Find the eigenvalues and the corresponding eigenvectors of the following matrix $A = \begin{pmatrix} 2 & 7 \\ -1 & -6 \end{pmatrix}$

The eigen values of A is obtained from

$$\begin{vmatrix} 2 - \lambda & 7 \\ -1 & -6 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (2 - \lambda)(-6 - \lambda) + 7 = 0$$

$$\Rightarrow \lambda^2 + 4\lambda - 5 = 0$$

$$\Rightarrow (\lambda + 5)(\lambda - 1) = 0$$

$$\Rightarrow \lambda = -5, 1,$$

For $\lambda = 5$, let $\eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$ be the corresponding eigenvector, then

$$\begin{pmatrix} 7 & 7 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow 7\eta_1 + 7\eta_2 = 0$$

$$\Rightarrow \eta_1 = -\eta_2$$

The eigenvector in this case $\eta = \begin{pmatrix} -\eta_2 \\ \eta_2 \end{pmatrix}$
 $\Rightarrow \eta^{(1)} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ for $\eta_2 = 1$

Again for $\lambda = 1$, let $\eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$ be the corresponding eigenvector, then

$$\begin{pmatrix} 1 & 7 \\ -1 & -7 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \eta_1 + 7\eta_2 = 0 \Rightarrow \eta_1 = -7\eta_2$$

The eigenvector in this case

$$\eta = \begin{pmatrix} -7 \\ 1 \end{pmatrix} \quad \text{for } \eta_2 = 1$$

Eigenvalues and Eigenvectors

Example 2 Find the eigenvalues and eigenvectors of the following matrix.

$$A = \begin{pmatrix} 1 & -1 \\ \frac{4}{9} & -\frac{1}{3} \end{pmatrix}$$

Solution

This matrix has fractions in it. That's life so don't get excited about it. First, we need the eigenvalues.

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 1 - \lambda & -1 \\ \frac{4}{9} & -\frac{1}{3} - \lambda \end{vmatrix} \\ &= (1 - \lambda) \left(-\frac{1}{3} - \lambda \right) + \frac{4}{9} \\ &= \lambda^2 - \frac{2}{3}\lambda + \frac{1}{9} \\ &= \left(\lambda - \frac{1}{3} \right)^2 \quad \Rightarrow \quad \lambda_{1,2} = \frac{1}{3} \end{aligned}$$

Eigenvalues and Eigenvectors

There is only one eigenvalue so let's do the work for that one. We will need to solve the following system,

$$\begin{pmatrix} \frac{2}{3} & -1 \\ \frac{4}{9} & -\frac{2}{3} \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \Rightarrow \quad R_1 = \frac{3}{2} R_2$$

So, the rows are multiples of each other. We'll work with the first equation in this example to find the eigenvector.

$$\frac{2}{3}\eta_1 - \eta_2 = 0 \qquad \eta_2 = \frac{2}{3}\eta_1$$

Recall in the last example we decided that we wanted to make these as “nice” as possible and so should avoid fractions if we can. Sometimes, as in this case, we simply can't so we'll have to deal with it. In this case the eigenvector will be,

$$\vec{\eta} = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \begin{pmatrix} \eta_1 \\ \frac{2}{3}\eta_1 \end{pmatrix}, \qquad \eta_1 \neq 0$$
$$\vec{\eta}^{(1)} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \qquad \eta_1 = 3$$

Eigenvalues and Eigenvectors

Example1: Find the eigenvalues and the corresponding eigenvectors of the following matrix

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

The eigen values of A is obtained from

$$\begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 + 3\lambda + 2 = 0$$

$$\Rightarrow (\lambda - 2)(\lambda + 1)^2 = 0$$

$$\Rightarrow \lambda = 2, -1, -1$$

For $\lambda = 2$, let $\eta = \begin{pmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{pmatrix}$ be the corresponding eigenvector, then

$$\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -2 & 1 & 1 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -2 & 0 \end{pmatrix} \xrightarrow{H_{12}} \begin{pmatrix} 1 & -2 & 1 & 0 \\ -2 & 1 & 1 & 0 \\ 1 & 1 & -2 & 0 \end{pmatrix} \xrightarrow{\substack{H_{21}(2) \\ H_{31}(-1)}} \begin{pmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 3 & -3 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 3 & -3 & 0 \end{pmatrix} \xrightarrow{H_2(-\frac{1}{3})} \begin{pmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 3 & -3 & 0 \end{pmatrix} \xrightarrow{H_{32}(-3)} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Going back to the equation gives

$$\begin{aligned} \eta_1 - \eta_3 &= 0 & \Rightarrow & \eta_1 = \eta_3 \\ \eta_2 - \eta_3 &= 0 & \Rightarrow & \eta_2 = \eta_3 \end{aligned}$$

$$\eta^{(1)} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad \text{For } \eta_3 = 1$$

Eigenvalues and Eigenvectors

$$\lambda_2 = -1 :$$

Here we'll need to solve,

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Going back to equations gives,

$$\eta_1 + \eta_2 + \eta_3 = 0 \Rightarrow \eta_1 = -\eta_2 - \eta_3$$

$$\vec{\eta} = \begin{pmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{pmatrix} = \begin{pmatrix} -\eta_2 - \eta_3 \\ \eta_2 \\ \eta_3 \end{pmatrix},$$

$\eta_2 \neq 0$ and $\eta_3 \neq 0$ at the same time

$$\vec{\eta}^{(2)} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \quad \eta_2 = 0 \text{ and } \eta_3 = 1$$

$$\vec{\eta}^{(3)} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \quad \eta_2 = 1 \text{ and } \eta_3 = 0$$

So, summarizing up,

$$\lambda_1 = 2 \quad \vec{\eta}^{(1)} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\lambda_2 = -1 \quad \vec{\eta}^{(2)} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$\lambda_3 = -1 \quad \vec{\eta}^{(3)} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$$

Solutions of System of DE's

The system of DE's are as follows

$$\begin{aligned}x'_1 &= a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\x'_2 &= a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\&\vdots \\x'_n &= a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n\end{aligned}$$

In matrix form above system can be written as

$$X' = AX \quad (1)$$

$$\text{where } A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ \vdots & & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \quad X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

If $n = 1$, the system reduces to a simple linear first order DE: $x' = ax$

and this has the solution $x = ce^{at}$

So, let's use this as a guide and for a general n let's consider

$$X = \eta e^{\lambda t} \quad (2)$$

Substituting Eq. (2) into Eq. (1), we obtain

$$\lambda \eta e^{\lambda t} = A \eta e^{\lambda t} \Rightarrow A \eta e^{\lambda t} - \lambda \eta e^{\lambda t} = 0 \Rightarrow (A - \lambda I) \eta e^{\lambda t} = 0$$

Since we know that exponentials are not zero, hence $(A - \lambda I) \eta = 0$

In order for (2) to be a solution to (1), λ and η must be an eigenvalue and eigenvector for the matrix A .

Therefore, in order to solve (1) we first find the eigenvalues and eigenvectors of the matrix A and then we can form solutions using (2).

Solutions of System of DE's (Real and Distinct Eigenvalues)

If λ_1 and λ_2 are two real and distinct eigenvalues, then the general solution in this case will be $X = \eta^{(1)} e^{\lambda_1 t} + \eta^{(2)} e^{\lambda_2 t}$

Example1: Solve the initial value problem

$$\begin{aligned} x'_1 &= x_1 + 2x_2 \\ x'_2 &= 3x_1 + 2x_2 \end{aligned} \quad \text{where } \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} 0 \\ -4 \end{pmatrix}$$

Solution:

In matrix form the above system can be written as

$$X' = AX \quad \text{Here } A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} \quad X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

The eigen values of A is obtained from

$$\begin{vmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)(2-\lambda) - 6 = 0 \Rightarrow \lambda^2 - 3\lambda - 40 = 0$$

$$\Rightarrow (\lambda + 1)(\lambda - 4) = 0$$

$$\Rightarrow \lambda = -1, 4$$

For $\lambda = -1$, let $\eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$ be the corresponding eigenvector, then

$$\begin{pmatrix} 2 & 2 \\ 3 & 3 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow 2\eta_1 + 2\eta_2 = 0 \Rightarrow \eta_1 = -\eta_2$$

The eigenvector in this case $\eta = \begin{pmatrix} -\eta_2 \\ \eta_2 \end{pmatrix}$

$$\Rightarrow \eta^{(1)} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad \text{for } \eta_2 = 1$$

Again for $\lambda = 4$, let $\eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$ be the corresponding eigenvector, then

$$\begin{pmatrix} -3 & 2 \\ 3 & -2 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow -3\eta_1 + 2\eta_2 = 0 \Rightarrow 3\eta_1 = 2\eta_2$$

The eigenvector in this case

$$\eta = \begin{pmatrix} \frac{2}{3}\eta_2 \\ \eta_2 \end{pmatrix} \Rightarrow \eta^{(2)} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad \text{for } \eta_2 = 3$$

Solutions of System of DE's (Real and Distinct Eigenvalues)

Then the general solution is

$$X = c_1 e^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + c_2 e^{4t} \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

Using initial condition, we get

$$\begin{pmatrix} 0 \\ -4 \end{pmatrix} = c_1 \begin{pmatrix} -1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

This gives

$$\begin{aligned} -c_1 + 2c_2 &= 0 \\ c_1 + 3c_2 &= -4 \end{aligned}$$

Solving above equations, we get

$$c_1 = -\frac{8}{5}, \quad c_2 = -\frac{4}{5}$$

$$X = -\frac{8}{5} e^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} - \frac{4}{5} e^{4t} \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$X = \frac{4}{5} \begin{pmatrix} 2e^{-t} - 2e^{4t} \\ -2e^{-t} - 3e^{4t} \end{pmatrix}$$

Example2: Solve the following initial value problem

$$X' = \begin{pmatrix} -5 & 1 \\ 4 & -2 \end{pmatrix} X \quad X(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Solution: The eigen values of A is obtained from

$$\begin{vmatrix} -5 - \lambda & 1 \\ 4 & -2 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (-5 - \lambda)(-2 - \lambda) - 4 = 0 \quad \Rightarrow \lambda^2 + 7\lambda + 6 = 0$$

$$\Rightarrow (\lambda + 1)(\lambda + 6) = 0 \quad \Rightarrow \lambda = -1, -6$$

For $\lambda = -1$, let $\eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$ be the corresponding eigenvector, then

$$\begin{pmatrix} -4 & 1 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow -4\eta_1 + \eta_2 = 0 \Rightarrow \eta_2 = 4\eta_1$$

The eigenvector in this case $\eta = \begin{pmatrix} \eta_1 \\ 4\eta_1 \end{pmatrix}$

$$\Rightarrow \eta^{(1)} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} \quad \text{for } \eta_1 = 1$$

Solutions of System of DE's (Real and Distinct Eigenvalues)

Again for $\lambda = -6$, let $\eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$ be the corresponding eigenvector, then

$$\begin{pmatrix} 1 & 1 \\ 4 & 4 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \eta_1 + \eta_2 = 0 \Rightarrow \eta_1 = -\eta_2$$

The eigenvector in this case

$$\eta = \begin{pmatrix} -\eta_2 \\ \eta_2 \end{pmatrix} \Rightarrow \eta^{(2)} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \text{ for } \eta_2 = 1$$

Then the general solution is

$$X = c_1 e^{-t} \begin{pmatrix} 1 \\ 4 \end{pmatrix} + c_2 e^{-6t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Using initial condition, we get $\begin{pmatrix} 1 \\ 2 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 4 \end{pmatrix} + c_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

This gives
$$\begin{cases} c_1 - c_2 = 1 \\ 4c_1 + c_2 = 2 \end{cases} \Rightarrow c_1 = \frac{3}{5}, c_2 = -\frac{2}{5}$$

$$X = \frac{3}{5} e^{-t} \begin{pmatrix} 1 \\ 4 \end{pmatrix} - \frac{2}{5} e^{-6t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Example3: Convert the following differential equation into a system, solve the system and use this solution to get the solution to the original differential equation

Solutions of System of DE's (Complex Eigenvalues)

Example1: Solve the initial value problem

$$\begin{aligned} x'_1 &= 3x_1 + 9x_2 \\ x'_2 &= -4x_1 - 3x_2 \end{aligned} \quad \text{where } \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$$

Solution:

In matrix form the above system can be written as

$$X' = AX \quad \text{Here } A = \begin{pmatrix} 3 & 9 \\ -4 & -3 \end{pmatrix} \quad X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

The eigen values of A is obtained from

$$\begin{vmatrix} 3 - \lambda & 9 \\ -4 & -3 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (3 - \lambda)(-3 - \lambda) + 36 = 0$$

$$\Rightarrow \lambda^2 + 27 = 0$$

$$\Rightarrow \lambda = \pm 3\sqrt{3}i$$

For $\lambda = 3\sqrt{3}i$, let $\eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$ be the corresponding eigenvector, then

$$\begin{pmatrix} 3 - 3\sqrt{3}i & 9 \\ -4 & -3 - 3\sqrt{3}i \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Using the first equation we get

$$\Rightarrow (3 - 3\sqrt{3}i)\eta_1 + 9\eta_2 = 0 \Rightarrow \eta_2 = -\frac{1}{3}(1 - \sqrt{3}i)\eta_1$$

The eigenvector in this case

$$\Rightarrow \eta^{(1)} = \begin{pmatrix} 3 \\ -1 + \sqrt{3}i \end{pmatrix} \quad \text{for } \eta_1 = 3$$

$$\text{Second eigenvector is } \eta^{(2)} = \begin{pmatrix} 3 \\ -1 - \sqrt{3}i \end{pmatrix}$$

$$x_1 = e^{3\sqrt{3}it} \begin{pmatrix} 3 \\ -1 + \sqrt{3}i \end{pmatrix}$$

$$x_1 = (\cos(3\sqrt{3}t) + i \sin(3\sqrt{3}t)) \begin{pmatrix} 3 \\ -1 + \sqrt{3}i \end{pmatrix}$$

Solutions of System of DE's (Complex Eigenvalues)

$$x_1 = \begin{pmatrix} 3\cos(3\sqrt{3}t) + 3i \sin(3\sqrt{3}t) \\ (-1 + \sqrt{3}i)(\cos(3\sqrt{3}t) + i \sin(3\sqrt{3}t)) \end{pmatrix} \Rightarrow x_1 = \begin{pmatrix} 3\cos(3\sqrt{3}t) + 3i \sin(3\sqrt{3}t) \\ -\cos(3\sqrt{3}t) - i \sin(3\sqrt{3}t) + \sqrt{3}i\cos(3\sqrt{3}t) - \sqrt{3} \sin(3\sqrt{3}t) \end{pmatrix}$$

$$x_1 = \begin{pmatrix} 3\cos(3\sqrt{3}t) \\ -\cos(3\sqrt{3}t) - \sqrt{3} \sin(3\sqrt{3}t) \end{pmatrix} + i \begin{pmatrix} 3 \sin(3\sqrt{3}t) \\ -\sin(3\sqrt{3}t) + \sqrt{3}\cos(3\sqrt{3}t) \end{pmatrix} = u(t) + iv(t)$$

The general solution of the system with complex root is

$$X = c_1 u(t) + c_2 v(t) = c_1 \begin{pmatrix} 3\cos(3\sqrt{3}t) \\ -\cos(3\sqrt{3}t) - \sqrt{3} \sin(3\sqrt{3}t) \end{pmatrix} + c_2 \begin{pmatrix} 3 \sin(3\sqrt{3}t) \\ -\sin(3\sqrt{3}t) + \sqrt{3}\cos(3\sqrt{3}t) \end{pmatrix}$$

Using initial condition, we get $\begin{pmatrix} 2 \\ -4 \end{pmatrix} = c_1 \begin{pmatrix} 3 \\ -1 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ \sqrt{3} \end{pmatrix} \Rightarrow \begin{cases} 3c_1 = 2 \\ -c_1 + \sqrt{3}c_2 = -4 \end{cases} \Rightarrow c_1 = \frac{2}{3}, c_2 = -\frac{10}{3\sqrt{3}}$

$$X = \frac{2}{3} \begin{pmatrix} 3\cos(3\sqrt{3}t) \\ -\cos(3\sqrt{3}t) - \sqrt{3} \sin(3\sqrt{3}t) \end{pmatrix} - \frac{10}{3\sqrt{3}} \begin{pmatrix} 3 \sin(3\sqrt{3}t) \\ -\sin(3\sqrt{3}t) + \sqrt{3}\cos(3\sqrt{3}t) \end{pmatrix}$$

Solutions of System of DE's (Complex Eigenvalues)

Example2: Solve the following initial value problem

$$X' = \begin{pmatrix} 3 & -13 \\ 5 & 1 \end{pmatrix} X \quad X(0) = \begin{pmatrix} 3 \\ -10 \end{pmatrix}$$

Solution: The eigen values of A is obtained from

$$\begin{aligned} \begin{vmatrix} 3 - \lambda & -13 \\ 5 & 1 - \lambda \end{vmatrix} &= 0 \\ \Rightarrow (3 - \lambda)(1 - \lambda) + 65 &= 0 \\ \Rightarrow \lambda^2 - 4\lambda + 68 &= 0 \\ \Rightarrow \lambda &= 2 \pm 8i \end{aligned}$$

For $\lambda = 2 + 8i$, let $\eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$ be the corresponding eigenvector, then

$$\begin{pmatrix} 1 - 8i & -13 \\ 5 & -1 - 8i \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Using second equation we get

$$\begin{aligned} \Rightarrow 5\eta_1 - (1 + 8i)\eta_2 &= 0 \\ \Rightarrow \eta_1 &= \frac{1}{5}(1 + 8i)\eta_2 \end{aligned}$$

The eigenvector in this case

$$\Rightarrow \eta^{(1)} = \begin{pmatrix} 1 + 8i \\ 5 \end{pmatrix} \quad \text{for } \eta_2 = 5$$

The solution corresponding to this eigenvalue and eigenvector is

$$x_1 = e^{(2+8i)t} \begin{pmatrix} 1 + 8i \\ 5 \end{pmatrix}$$

$$x_1 = e^{2t}(\cos(8t) + i \sin(8t)) \begin{pmatrix} 1 + 8i \\ 5 \end{pmatrix}$$

$$x_1 = e^{2t} \begin{pmatrix} \cos(8t) + i \sin(8t) + 8i \cos(8t) - 8 \sin(8t) \\ 5\cos(8t) + 5i \sin(8t) \end{pmatrix}$$

Solutions of System of DE's (Complex Eigenvalues)

$$x_1 = e^{2t} \begin{pmatrix} \cos(8t) - 8 \sin(8t) \\ 5 \cos(8t) \end{pmatrix} + i e^{2t} \begin{pmatrix} \sin(8t) + 8 \cos(8t) \\ 5 \sin(8t) \end{pmatrix} = u(t) + iv(t)$$

The general solution of the system with complex root is

$$X = c_1 e^{2t} \begin{pmatrix} \cos(8t) - 8 \sin(8t) \\ 5 \cos(8t) \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} \sin(8t) + 8 \cos(8t) \\ 5 \sin(8t) \end{pmatrix}$$

Using initial condition, we get

$$\begin{pmatrix} 3 \\ -10 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 5 \end{pmatrix} + c_2 \begin{pmatrix} 8 \\ 0 \end{pmatrix} \quad \longrightarrow \quad \begin{matrix} c_1 + 8c_2 = 3 \\ 5c_1 = -10 \end{matrix} \quad \Rightarrow c_1 = -2, c_2 = \frac{5}{8}$$

$$X = -2e^{2t} \begin{pmatrix} \cos(8t) - 8 \sin(8t) \\ 5 \cos(8t) \end{pmatrix} + \frac{5}{8}e^{2t} \begin{pmatrix} \sin(8t) + 8 \cos(8t) \\ 5 \sin(8t) \end{pmatrix}$$

Solutions of System of DE's (Repeated Eigenvalues)

In matrix form, consider a system DE's as $x' = Ax$ (1)

where the eigenvalues are repeated eigenvalues. Since we are going to be working with systems in which A is a 2×2 matrix we will make that assumption from the start. So, the system will have a double eigenvalue, λ .

We want two linearly independent solutions so that we can form a general solution. However, with a double eigenvalue we will have only one, $x_1 = \eta e^{\lambda t}$

So, we need to come up with a second solution. Recall that when we looked at the double root case with the second order differential equations we ran into a similar problem. In that section we simply added a t to the solution and were able to get a second solution. Let's see if the same thing will work in this case as well. We'll see if

$$x = \eta t e^{\lambda t} \quad (2) \quad \text{will also be a solution}$$

Substituting Eq.(2) into Eq.(1), we get $\eta \lambda t e^{\lambda t} + \eta e^{\lambda t} = A \eta t e^{\lambda t}$

$$\text{Equating the coefficient of like terms we get} \quad \eta \lambda = A \eta \quad \Rightarrow (A - \lambda I) \eta = 0$$

$$\eta = 0$$

The first requirement isn't a problem since this just says that λ is an eigenvalue and its eigenvector is η .

The second however is a problem. Since η is an eigenvector we know that it can't be zero, yet in order to satisfy the second condition it would have to be.

Solutions of System of DE's (Repeated Eigenvalues)

Let's try the following guess $x = \eta t e^{\lambda t} + \rho e^{\lambda t}$ (3) where ρ is an unknown vector that we'll need to determine

Substituting Eq.(3) into Eq.(1), we get $\eta \lambda t e^{\lambda t} + \rho \lambda e^{\lambda t} + \eta e^{\lambda t} = A \eta t e^{\lambda t} + A \rho e^{\lambda t}$

Equating the coefficient of like terms we get $\eta \lambda = A \eta \Rightarrow (A - \lambda I) \eta = 0$
 $\rho \lambda + \eta = A \rho \Rightarrow (A - \lambda I) \rho = \eta$

This time the second equation is not a problem. The second equation tells us that ρ must be a solution to this equation.

It looks like our second guess worked. Therefore $x_2 = \eta t e^{\lambda t} + \rho e^{\lambda t}$

will be a solution to the system provided ρ is a solution to $(A - \lambda I) \rho = \eta$

So the general solution in the double eigenvalue case is $x = c_1 x_1 + c_2 x_2 = c_1 \eta e^{\lambda t} + c_2 (\eta t e^{\lambda t} + \rho e^{\lambda t})$

Solutions of System of DE's (Repeated Eigenvalues)

Example2: Solve the following initial value problem

$$X' = \begin{pmatrix} 7 & 1 \\ -4 & 3 \end{pmatrix} X \quad X(0) = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$$

Solution: The eigen values of A is obtained from

$$\begin{aligned} \begin{vmatrix} 7-\lambda & 1 \\ -4 & 3-\lambda \end{vmatrix} &= 0 \\ \Rightarrow (3-\lambda)(7-\lambda) + 4 &= 0 \\ \Rightarrow \lambda^2 - 10\lambda + 25 &= 0 \\ \Rightarrow \lambda &= 5, 5 \end{aligned}$$

For $\lambda = 5$, let $\eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$ be the corresponding eigenvector, then

$$\begin{pmatrix} 2 & 1 \\ -4 & -2 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow 2\eta_1 + \eta_2 = 0 \Rightarrow \eta_2 = -2\eta_1$$

The eigenvector in this case

$$\Rightarrow \eta = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad \text{for } \eta_1 = 1$$

To find ρ , we need to solve

$$\begin{pmatrix} 2 & 1 \\ -4 & -2 \end{pmatrix} \begin{pmatrix} \rho_1 \\ \rho_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \Rightarrow 2\rho_1 + \rho_2 = 1 \Rightarrow \rho_2 = 1 - 2\rho_1$$

$$\text{Therefore } \rho = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \text{if } \rho_1 = 0$$

The general solution is

$$X = c_1 e^{5t} \begin{pmatrix} 1 \\ -2 \end{pmatrix} + c_2 (t e^{5t} \begin{pmatrix} 1 \\ -2 \end{pmatrix} + e^{5t} \begin{pmatrix} 0 \\ 1 \end{pmatrix})$$

Using initial condition, we get

$$\begin{aligned} \begin{pmatrix} 2 \\ -5 \end{pmatrix} &= c_1 \begin{pmatrix} 1 \\ -2 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ c_1 &= 2 \Rightarrow c_1 = 2, c_2 = -1 \\ -2c_1 + c_2 &= -5 \end{aligned}$$

$$X = 2e^{5t} \begin{pmatrix} 1 \\ -2 \end{pmatrix} - t e^{5t} \begin{pmatrix} 1 \\ -2 \end{pmatrix} - e^{5t} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$X = 2e^{5t} \begin{pmatrix} 2-t \\ -5+2t \end{pmatrix}$$

Solutions of System of DE's (Repeated Eigenvalues)

Example 3 Solve the following IVP.

$$\vec{x}' = \begin{pmatrix} -1 & \frac{3}{2} \\ -\frac{1}{6} & -2 \end{pmatrix} \vec{x} \quad \vec{x}(2) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Solution

First the eigenvalue for the system.

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} -1-\lambda & \frac{3}{2} \\ -\frac{1}{6} & -2-\lambda \end{vmatrix} \\ &= \lambda^2 + 3\lambda + \frac{9}{4} \\ &= \left(\lambda + \frac{3}{2}\right)^2 \quad \Rightarrow \quad \lambda_{1,2} = -\frac{3}{2} \end{aligned}$$

Now let's get the eigenvector.

$$\begin{aligned} \begin{pmatrix} \frac{1}{2} & \frac{3}{2} \\ -\frac{1}{6} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \Rightarrow \quad \frac{1}{2}\eta_1 + \frac{3}{2}\eta_2 = 0 \quad \eta_1 = -3\eta_2 \\ \vec{\eta} &= \begin{pmatrix} -3\eta_2 \\ \eta_2 \end{pmatrix} \quad \eta_2 \neq 0 \\ \vec{\eta}^{(1)} &= \begin{pmatrix} -3 \\ 1 \end{pmatrix} \quad \eta_2 = 1 \end{aligned}$$

Now find $\vec{\rho}$,

$$\begin{aligned} \begin{pmatrix} \frac{1}{2} & \frac{3}{2} \\ -\frac{1}{6} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} \rho_1 \\ \rho_2 \end{pmatrix} &= \begin{pmatrix} -3 \\ 1 \end{pmatrix} \quad \Rightarrow \quad \frac{1}{2}\rho_1 + \frac{3}{2}\rho_2 = -3 \quad \rho_1 = -6 - 3\rho_2 \\ \vec{\rho} &= \begin{pmatrix} -6 - 3\rho_2 \\ \rho_2 \end{pmatrix} \quad \Rightarrow \quad \vec{\rho} = \begin{pmatrix} -6 \\ 0 \end{pmatrix} \quad \text{if } \rho_2 = 0 \end{aligned}$$

$$\vec{x}(t) = c_1 \mathbf{e}^{-\frac{3t}{2}} \begin{pmatrix} -3 \\ 1 \end{pmatrix} + c_2 \left(t \mathbf{e}^{-\frac{3t}{2}} \begin{pmatrix} -3 \\ 1 \end{pmatrix} + \mathbf{e}^{-\frac{3t}{2}} \begin{pmatrix} -6 \\ 0 \end{pmatrix} \right)$$

Applying the initial condition gives,

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \vec{x}(2) = c_1 \mathbf{e}^{-3} \begin{pmatrix} -3 \\ 1 \end{pmatrix} + c_2 \left(2\mathbf{e}^{-3} \begin{pmatrix} -3 \\ 1 \end{pmatrix} + \mathbf{e}^{-3} \begin{pmatrix} -6 \\ 0 \end{pmatrix} \right)$$

Note that we didn't use $t=0$ this time! We now need to solve the following system,

$$\begin{cases} -3\mathbf{e}^{-3}c_1 - 12\mathbf{e}^{-3}c_2 = 1 \\ \mathbf{e}^{-3}c_1 + 2\mathbf{e}^{-3}c_2 = 0 \end{cases} \quad \Rightarrow \quad c_1 = \frac{\mathbf{e}^3}{3}, \quad c_2 = -\frac{\mathbf{e}^3}{6}$$

The actual solution is then,

$$\begin{aligned} \vec{x}(t) &= \frac{\mathbf{e}^3}{3} \mathbf{e}^{-\frac{3t}{2}} \begin{pmatrix} -3 \\ 1 \end{pmatrix} - \frac{\mathbf{e}^3}{6} \left(t \mathbf{e}^{-\frac{3t}{2}} \begin{pmatrix} -3 \\ 1 \end{pmatrix} + \mathbf{e}^{-\frac{3t}{2}} \begin{pmatrix} -6 \\ 0 \end{pmatrix} \right) \\ &= \mathbf{e}^{-\frac{3t}{2}+3} \begin{pmatrix} 0 \\ \frac{1}{3} \end{pmatrix} + t \mathbf{e}^{-\frac{3t}{2}+3} \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{6} \end{pmatrix} \end{aligned}$$

Vector Space

Vector Space:

DEFINITION: Let V be a nonempty set with two operations:

- (i) *Vector Addition:* This assigns to any $u, v \in V$ a sum $u + v$ in V .
- (ii) *Scalar Multiplication:* This assigns to any $u \in V, k \in K$ a product $ku \in V$.

Then V is called a *vector space* (over the field K) if the following axioms hold for any vectors $u, v, w \in V$:

[A₁] $(u + v) + w = u + (v + w)$

[A₂] There is a vector in V , denoted by 0 and called the *zero vector*, such that, for any $u \in V$,

$$u + 0 = 0 + u = u$$

[A₃] For each $u \in V$, there is a vector in V , denoted by $-u$, and called the *negative* of u , such that

$$u + (-u) = (-u) + u = 0.$$

[A₄] $u + v = v + u$.

[M₁] $k(u + v) = ku + kv$, for any scalar $k \in K$.

[M₂] $(a + b)u = au + bu$, for any scalars $a, b \in K$.

[M₃] $(ab)u = a(bu)$, for any scalars $a, b \in K$.

[M₄] $1u = u$, for the unit scalar $1 \in K$.

Vector Space

4.4 Linear Combinations, Spanning Sets

Let V be a vector space over a field K . A vector v in V is a *linear combination* of vectors $u_1, u_2, \dots, u_m$ in V if there exist scalars $a_1, a_2, \dots, a_m$ in K such that

$$v = a_1 u_1 + a_2 u_2 + \cdots + a_m u_m$$

Alternatively, v is a linear combination of $u_1, u_2, \dots, u_m$ if there is a solution to the vector equation

$$v = x_1 u_1 + x_2 u_2 + \cdots + x_m u_m$$

where $x_1, x_2, \dots, x_m$ are unknown scalars.

Vector Space

EXAMPLE 4.1 (Linear Combinations in $\mathbf{R}^n$) Suppose we want to express $v = (3, 7, -4)$ in $\mathbf{R}^3$ as a linear combination of the vectors

$$u_1 = (1, 2, 3), \quad u_2 = (2, 3, 7), \quad u_3 = (3, 5, 6)$$

We seek scalars x, y, z such that $v = xu_1 + yu_2 + zu_3$; that is,

$$\begin{bmatrix} 3 \\ 7 \\ -4 \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + y \begin{bmatrix} 2 \\ 3 \\ 7 \end{bmatrix} + z \begin{bmatrix} 3 \\ 5 \\ 6 \end{bmatrix} \quad \text{or} \quad \begin{aligned} x + 2y + 3z &= 3 \\ 2x + 3y + 5z &= 7 \\ 3x + 7y + 6z &= -4 \end{aligned}$$

(For notational convenience, we have written the vectors in $\mathbf{R}^3$ as columns, because it is then easier to find the equivalent system of linear equations.) Reducing the system to echelon form yields

$$\begin{array}{ll} x + 2y + 3z = 3 & x + 2y + 3z = 3 \\ -y - z = 1 & \text{and then} \quad -y - z = 1 \\ y - 3z = -13 & -4z = -12 \end{array}$$

Back-substitution yields the solution $x = 2, \quad y = -4, \quad z = 3$. Thus, $v = 2u_1 - 4u_2 + 3u_3$.

Spanning Sets

Let V be a vector space over K . Vectors $u_1, u_2, \dots, u_m$ in V are said to *span* V or to form a *spanning set* of V if every v in V is a linear combination of the vectors $u_1, u_2, \dots, u_m$ —that is, if there exist scalars $a_1, a_2, \dots, a_m$ in K such that

$$v = a_1u_1 + a_2u_2 + \cdots + a_mu_m$$

EXAMPLE 4.3 Consider the vector space $V = \mathbf{R}^3$.

(a) We claim that the following vectors form a spanning set of $\mathbf{R}^3$:

$$e_1 = (1, 0, 0), \quad e_2 = (0, 1, 0), \quad e_3 = (0, 0, 1)$$

Specifically, if $v = (a, b, c)$ is any vector in $\mathbf{R}^3$, then

$$v = ae_1 + be_2 + ce_3$$

For example, $v = (5, -6, 2) = -5e_1 - 6e_2 + 2e_3$.

(b) We claim that the following vectors also form a spanning set of $\mathbf{R}^3$:

$$w_1 = (1, 1, 1), \quad w_2 = (1, 1, 0), \quad w_3 = (1, 0, 0)$$

Specifically, if $v = (a, b, c)$ is any vector in $\mathbf{R}^3$, then (Problem 4.62)

$$v = (a, b, c) = cw_1 + (b - c)w_2 + (a - b)w_3$$

For example, $v = (5, -6, 2) = 2w_1 - 8w_2 + 11w_3$.

(c) One can show (Problem 3.24) that $v = (2, 7, 8)$ cannot be written as a linear combination of the vectors

$$u_1 = (1, 2, 3), \quad u_2 = (1, 3, 5), \quad u_3 = (1, 5, 9)$$

Accordingly, u_1, u_2, u_3 do not span $\mathbf{R}^3$.

DEFINITION: We say that the vectors $v_1, v_2, \dots, v_m$ in V are *linearly dependent* if there exist scalars $a_1, a_2, \dots, a_m$ in K , not all of them 0, such that

$$a_1 v_1 + a_2 v_2 + \cdots + a_m v_m = 0$$

Otherwise, we say that the vectors are *linearly independent*.

(a) Let $u = (1, 1, 0)$, $v = (1, 3, 2)$, $w = (4, 9, 5)$. Then u , v , w are linearly dependent, because

$$3u + 5v - 2w = 3(1, 1, 0) + 5(1, 3, 2) - 2(4, 9, 5) = (0, 0, 0) = 0$$

(b) We show that the vectors $u = (1, 2, 3)$, $v = (2, 5, 7)$, $w = (1, 3, 5)$ are linearly independent. We form the vector equation $xu + yv + zw = 0$, where x, y, z are unknown scalars. This yields

$$x \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + y \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix} + z \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{or} \quad \begin{array}{l} x + 2y + z = 0 \\ 2x + 5y + 3z = 0 \\ 3x + 7y + 5z = 0 \end{array} \quad \text{or} \quad \begin{array}{l} x + 2y + z = 0 \\ y + z = 0 \\ 2z = 0 \end{array}$$

Back-substitution yields $x = 0$, $y = 0$, $z = 0$. We have shown that

$$xu + yv + zw = 0 \quad \text{implies} \quad x = 0, \quad y = 0, \quad z = 0$$